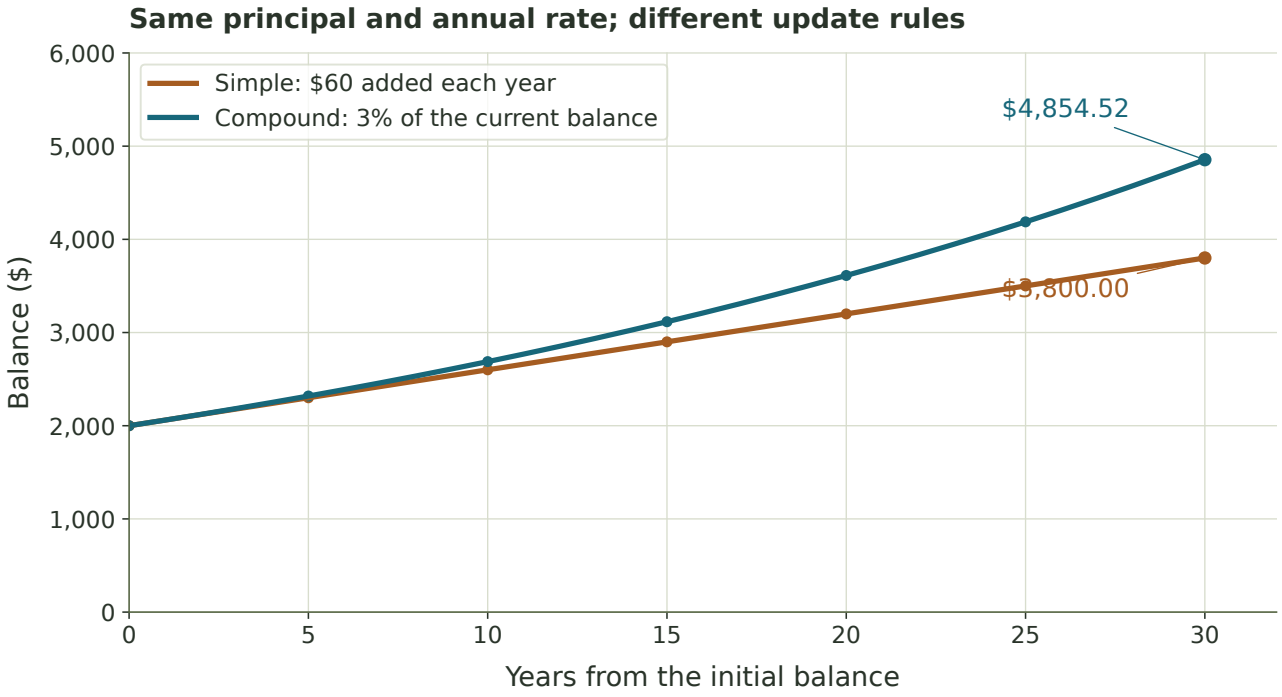
WEEK 6 / CONTINUOUS MATH I / INTEREST AND LOANS

Following the balance

Start with one amount. State the period and the rule. Use the ending balance as the next input.



The setting for the week

A $2,000 balance can grow through fixed dollar additions or a percentage of its current amount. A loan combines percentage growth with payments that subtract from the debt. Tables and graphs make the differences visible.

This chapter develops the meaning of percentages, period rates, and repeated updates before using formulas. It distinguishes an ending balance from interest earned, a scheduled payment from principal reduction, and a simulation's stopping month from an actual payoff.

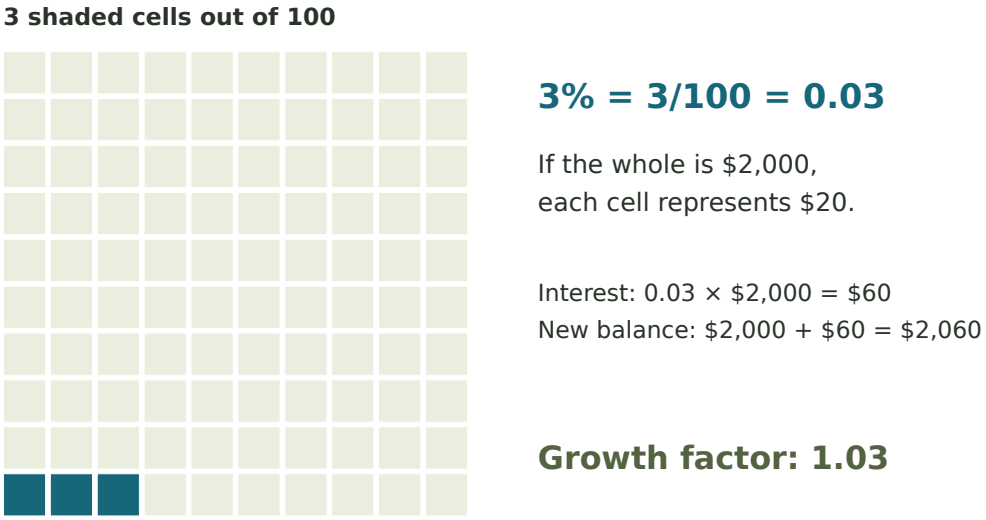
The core work uses arithmetic, powers, and simple equations. Try each short check before consulting its explanation. The review includes complete solutions and optional investigations of rounding and modeling assumptions.

Online reading · Interest and loan explorer · Guided laboratory

6.1 A rate needs an amount to act on and a period to describe

Principal means the initial amount deposited or borrowed. We use P for that starting amount and B for a balance as it changes. A *period* is one update interval, such as a year or month. Year 0 records the initial balance, before any annual interest is added.

A percentage means "per hundred." To calculate 3% of \$2,000, first write $3\% = 3/100 = 0.03$, then multiply: $0.03(2000) = 60$ dollars. To keep the original money and add that interest, multiply by $1 + 0.03 = 1.03$.



The rate 0.03 gives the interest portion. The factor 1.03 keeps the original balance and adds 3% to it.

Expression	Meaning	For a \$2,000 balance
0.03B	3% of the balance	Interest of \$60
B + 0.03B	Keep the balance and add its interest	New balance \$2,060
1.03B	The same update written as multiplication	New balance \$2,060

"3% per year" and "3% per month" describe different updates. In a formula, use a decimal rate whose time unit matches the period. Also keep separate the *interest amount* and the *total balance*: earning \$60 does not mean the whole account is worth \$60.

Example · Read the notation before calculating

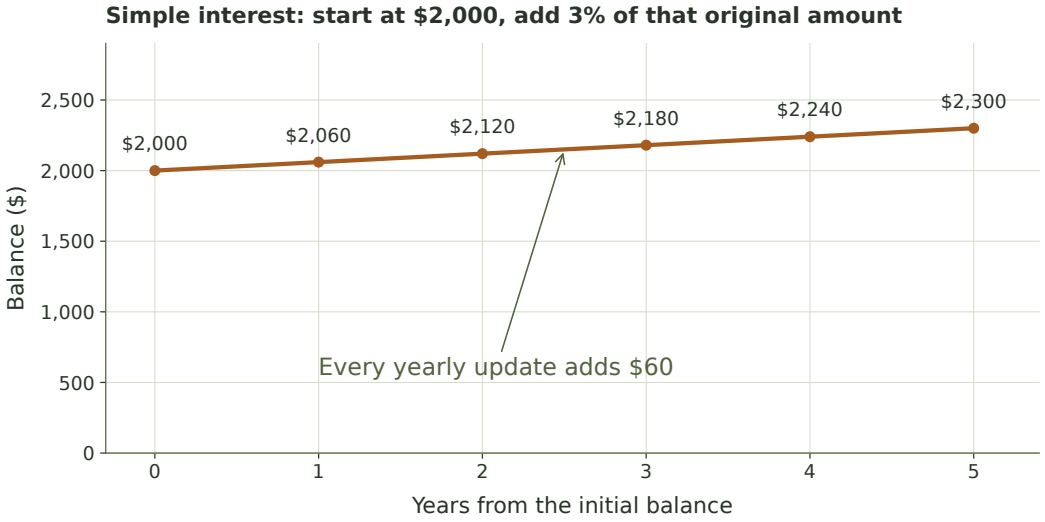
$B_0 = 2000$ names the starting balance. B_1 means the balance after one update, and B_2 after two. The small number is a period label, not a multiplier. The rule $B_{k+1} = 1.03B_k$ says "multiply the current balance by 1.03 to get the next one."

Pause and try 1

A \$500 balance earns 4% for one period. What is the interest, the growth factor, and the new balance?

6.2 Simple interest adds the same dollar amount each year

With *simple interest*, interest is calculated on the original principal, not on the accumulated balance. Start at P=$2,000 with annual rate r=0.03. Each year adds the same $Pr = 60$ dollars.



The equal dollar steps make a straight-line pattern. The year-5 balance includes the original $2,000 plus five $60 additions.

After t years, interest is $I = Prt$, and total amount is

$$A = P + Prt = P(1 + rt).$$

The formula collects the repeated additions into one calculation. With annual updates, we can also write $B_{k+1} = B_k + Pr$: the constant step Pr is added to the current total, but calculated from the *original* principal.

Example · Evaluate and solve backward

After five years, $A = 2000(1 + 0.03 \cdot 5) = 2300$ dollars. The interest earned is $2300 - 2000 = 300$ dollars.
Suppose instead a simple-interest account grows from $500 to $560 in four years. The $60 gain is $15 per year. Dividing $15 by the $500 original amount gives $r = 0.03$, or 3% per year. Algebra gives the same answer: $560/500 - 1 = 4r$, so $r = 0.12/4 = 0.03$.

If time is given in months while r is annual, convert months to years. Under a proportional simple-interest model, six months is t=0.5 year; $2,000 at 3% earns $2000(0.03)(0.5) = 30$ dollars over that time.

Pause and try 2

An account starts at $900 and earns 4% simple annual interest. What is the fixed yearly increase and the total after three years?

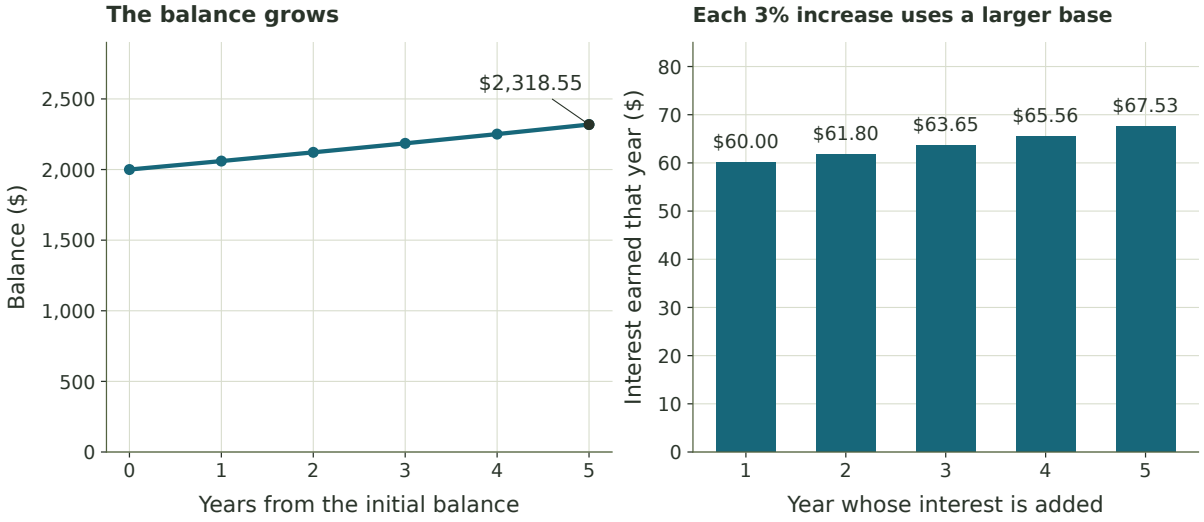
6.3 Compound interest lets earlier interest earn interest

With *compound interest*, each update calculates interest on the balance reached so far. At 3% annually, \$2,000 becomes \$2,060 after one year. The second year's interest is $0.03(2060) = 61.80$, so the next balance is \$2,121.80.

The annual rule is $B_{k+1} = (1 + r)B_k$. Applying it repeatedly gives

$$B_1 = P(1 + r), \quad B_2 = P(1 + r)^2, \quad A = P(1 + r)^t.$$

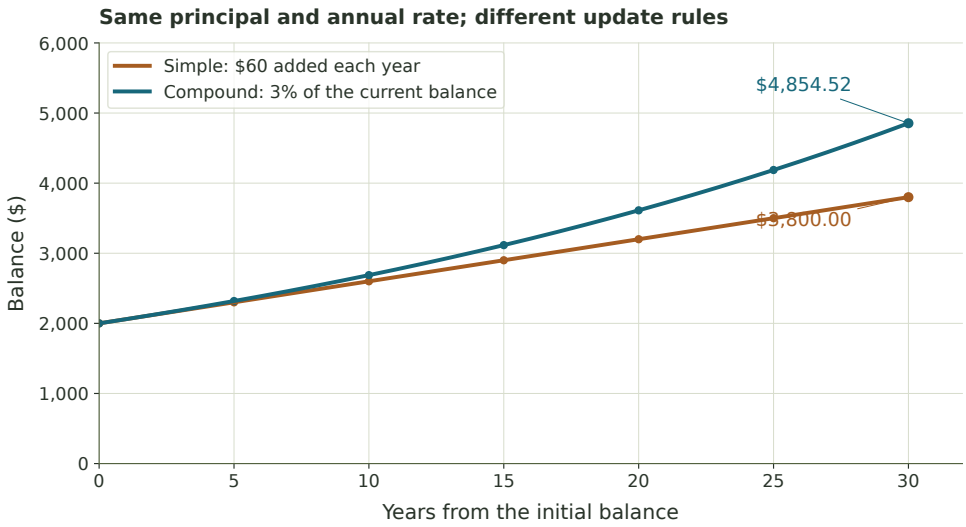
The exponent counts repeated multiplications. For three years, multiply by 1.03 three times; do not replace this by adding 3% of the original amount three times.



The percentage stays at 3%, while its dollar base grows. Table and chart values round for display; the compound formula keeps precision until the final answer.

Year	Simple balance (\$)	Annual compound balance (\$)
0	2,000.00	2,000.00
1	2,060.00	2,060.00
2	2,120.00	2,121.80
3	2,180.00	2,185.45
4	2,240.00	2,251.02
5	2,300.00	2,318.55

At the same positive annual rate, the two models agree after the first annual update. Later, compounding uses the earlier interest as part of its base. At 30 years, the simple model gives \$3,800, while the annual compound model gives about \$4,854.52.



Dots record balances at annual updates; connecting lines help compare the patterns. A gently increasing percentage process can accumulate a substantial difference over many periods.

A curve and a schedule are different parts of a model

The balance table records specified crediting dates. A continuous curve can help us visualize the growth pattern between them, but the account's terms determine when interest is actually credited. Our explorer uses whole years and specified compounding periods, so the displayed endpoints are completed updates.

Pause and try 3

Start with \$300 at 10% compounded annually. Find the first and second yearly balances, and compare with two years of simple interest.

6.4 Monthly compounding changes both the rate and the count

Suppose a model quotes a nominal annual rate r=0.06 and compounds 12 times a year. The rate per monthly update is $i = r/12 = 0.005$, which is 0.5%, not 6%. Over ten years there are $12(10) = 120$ monthly updates.

For n equal compounding periods per year and t years, the formula is

$$A = P\left(1 + \frac{r}{n}\right)^{nt}.$$

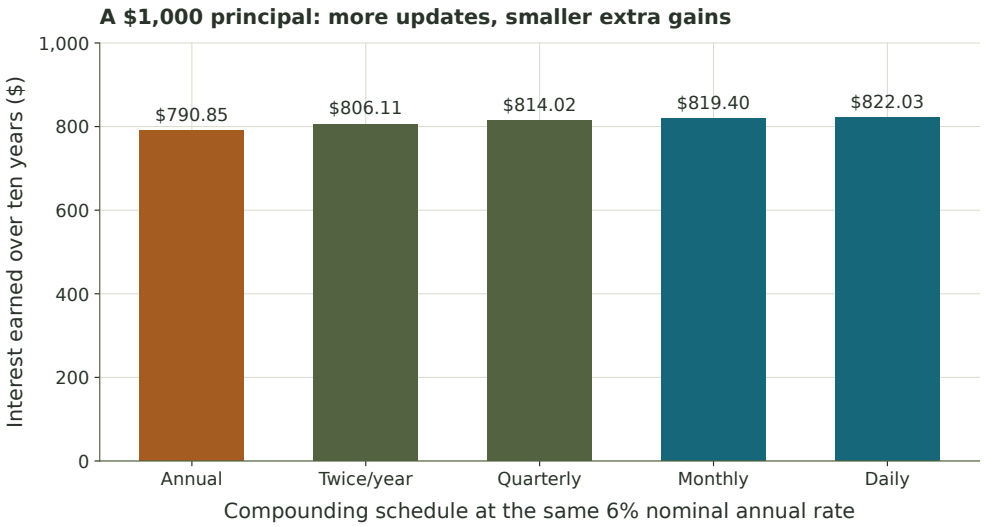
Read the inside and outside together: r/n is the rate for one step, and nt is the number of steps. Using a monthly rate with an annual count would model a different amount of time.

Example · Write the substituted expression first

For $1,000 at 6% nominal annual interest compounded monthly for ten years:

$$A = 1000\left(1 + \frac{0.06}{12}\right)^{120} = 1000(1.005)^{120} \approx 1819.40.$$

Use parentheses around the full factor and keep the calculator's precision through the exponent. The final interest earned is approximately $819.40.



The bars show interest earned, not the whole account balance. The comparison holds the nominal annual rate fixed. Daily means 365 equal periods in this constructed model.

For these positive rates, more frequent compounding increases the final amount, with smaller extra gains as the frequencies increase. This comparison does not hold the effective annual yield fixed.

The effective annual yield describes a whole year's growth

The *effective annual yield* is the fraction gained over one year, including compounding. For a 3% nominal annual rate compounded monthly,

$$\left(1+\frac{0.03}{12}\right)^{12}-1\approx 0.030416,$$

or about 3.0416%. The decimal 0.030416 and the percentage 3.0416% describe the same yield. The nominal rate 3% and the effective yield answer different questions about the update rule.

Pause and try 4

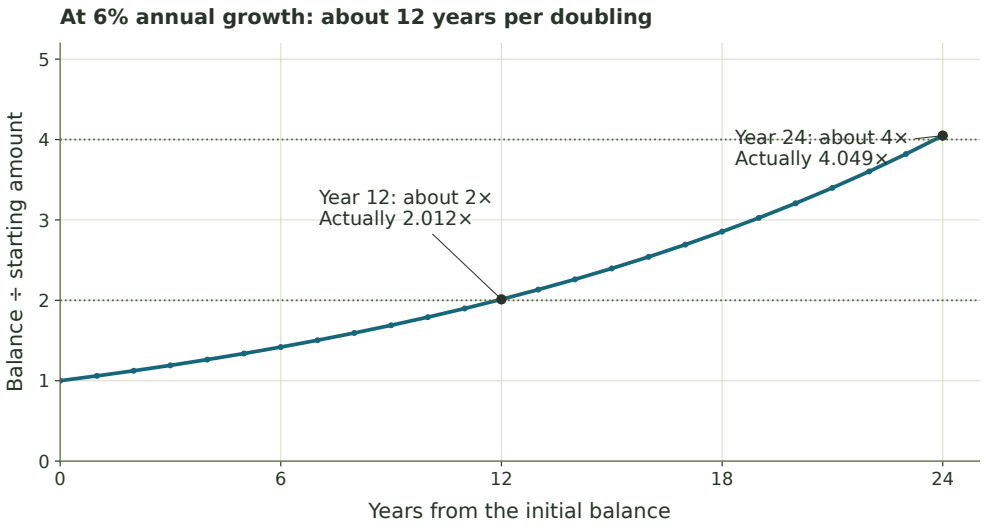
For a nominal annual rate of 12%, compounded monthly for two years, give the decimal monthly rate and the number of updates.

6.5 The Rule of 72 estimates repeated percentage growth

For a positive balance growing at a fixed positive annual compound rate, the *Rule of 72* estimates the years needed to double:

$$\text{doubling time}\approx\frac{72}{\text{annual rate written as a percentage number}}.$$

At 6%, use 72/6=12 years. Do not divide by 0.06. At 9%, the estimate is eight years. These are estimates under a continuing growth rule, not guaranteed future returns.



About 12 years produces one doubling, and about 24 produces two doublings. Two doublings multiply the starting amount by four, not by three.

Example · Check the estimate using powers

At 6% annual compounding, $(1.06)^{11}\approx 1.898$ and $(1.06)^{12}\approx 2.012$. The first completed annual update reaching twice the starting balance is year 12. The Rule of 72 is a useful scale estimate; it is not an equation making the balance exactly double at year 12.

The same reasoning works for invented compute capacity or another quantity following a fixed percentage rule. It does not describe simple interest, and it does not by itself describe a loan being reduced by payments. At a zero growth rate, a positive starting amount never doubles through this rule.

Pause and try 5

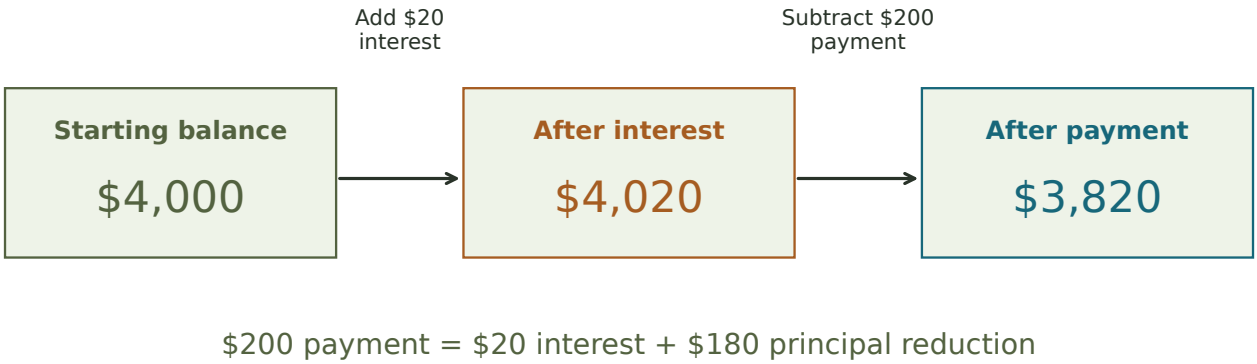
Use the Rule of 72 to estimate doubling time at 8% and quadrupling time at 6%.

6.6 A payment subtracts from a balance that has just grown

In our loan model, interest is added once a month, then the scheduled payment is applied. Let B be the balance at the start of that month, i the monthly decimal rate, and M the regular payment in dollars.

$$\text{interest} = iB, \qquad B_{\text{next}} = B + iB - M = (1+i)B - M.$$

The annual rate r is divided by 12 *under the stated monthly model*, so i=r/12. A \$4,000 loan at a 6% nominal annual rate uses i=0.06/12=0.005, or 0.5% per month. The first interest charge is \$20. In credit-card terminology, APR stands for *annual percentage rate*; our card exercises explicitly specify the monthly conversion used in their simplified model.



The \$200 payment covers \$20 interest and reduces principal by \$180. The ending balance \$3,820 becomes the next month's starting balance.

Example · Repeat using the new balance

Month 1: $4000(1.005) - 200 = 3820$. Month 2 starts at \$3,820, so its interest is $3820(0.005) = 19.10$ dollars. After the next \$200 payment, the balance is $3820 + 19.10 - 200 = 3639.10$.
Using \$4,000 again in month 2 would incorrectly keep charging the original interest amount. The current balance is part of the rule.

The credit-card examples use a simplified schedule

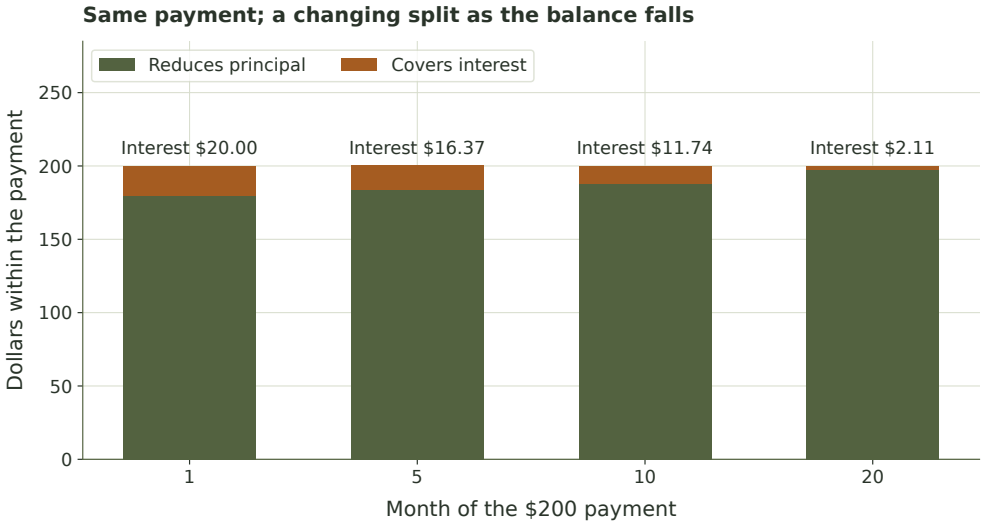
Our activities use fixed rates, no new purchases or fees, and one interest calculation followed by one payment each month. The Consumer Financial Protection Bureau explains that many real card issuers use daily balance calculations. The monthly model makes the repeated arithmetic visible; actual statements follow their account terms.

Pause and try 6

A $500 loan balance has 2% interest for one period, then a $60 payment. Find the interest, principal reduction, and ending balance.

6.7 Interest and principal reduction add up to the actual payment

For the decreasing loan above, the monthly payment stays $200, but its parts change. A smaller starting balance generates less interest. More of the same payment can then reduce what is owed.



The whole bar is the payment. Its interest portion is a cost of borrowing for that month; the remainder reduces the debt. These bars come from the $4,000, 6%, $200-payment model.

Month	Start balance ($)	Interest ($)	Payment ($)	End balance ($)
1	4,000.00	20.00	200.00	3,820.00
2	3,820.00	19.10	200.00	3,639.10
3	3,639.10	18.20	200.00	3,457.30
4	3,457.30	17.29	200.00	3,274.59

A table of this repeated repayment calculation is often called an *amortization schedule*. Check each row by adding interest to the start balance and subtracting the actual payment. The next row must begin with the previous row's ending balance.

State the rounding rule

The loan table in this reading rounds each month's interest to the nearest cent before updating the balance. The application's long comparison keeps full internal precision and rounds only the displayed results, preserving its original calculation method. Both conventions are available in the explorer. Small total differences can come from rounding rather than from a different interest formula.

Pause and try 7

In a month with \$16.37 interest and a \$200 actual payment, how much reduces principal? What would have to change for the next month's interest to be smaller?

6.8 The final payment is the amount still due

Continue the \$4,000 loan at 6% with \$200 monthly payments, rounding monthly interest to cents. After 21 regular payments, \$24.87 remains. Month 22 adds \$0.12 interest, so the final amount due is \$24.99. Paying that amount closes the loan.

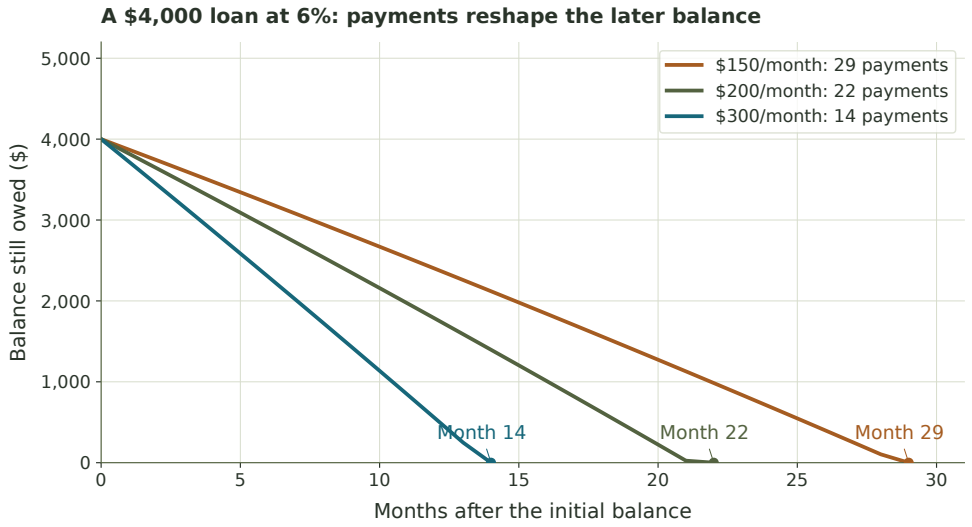
We do not subtract the full scheduled \$200 and report a negative debt. The actual last payment is the smaller of the regular payment and the amount due.

Example · Total paid and total interest

$$\text{total paid} = 21(200) + 24.99 = 4224.99,$$

$$\text{total interest} = 4224.99 - 4000 = 224.99.$$

The total paid includes the \$4,000 originally borrowed. The amount above that principal is the interest in this model, since there are no other fees or charges.



The curves share the same starting balance and first-month interest rate. Different payments create different later balances, which change later interest.

Regular payment	Payoff month	Total paid (\$)	Total interest (\$)
\$150	29	4,303.84	303.84
\$200	22	4,224.99	224.99
\$300	14	4,150.01	150.01

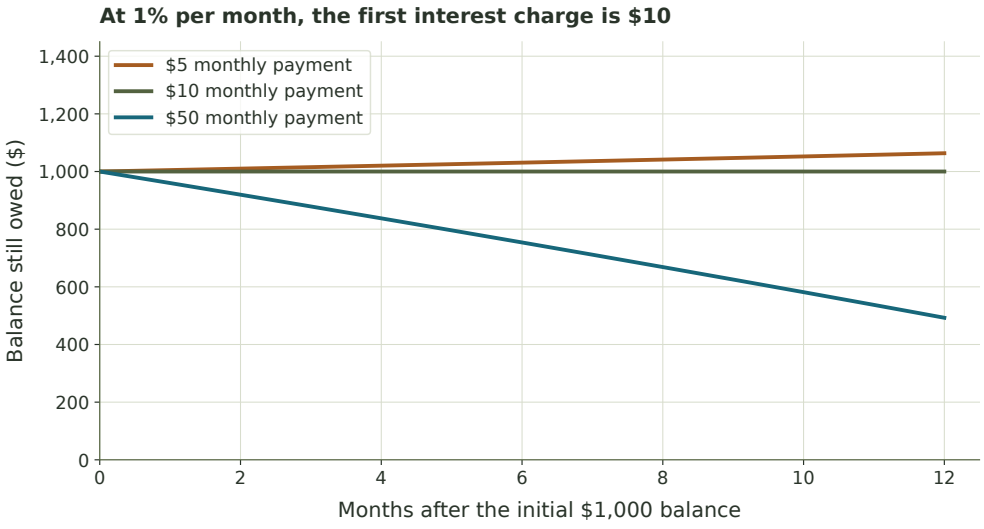
For these otherwise identical fixed-rate loans, larger payments reduce the balance faster and leave less balance on which later interest is charged. They also require more money each month. The table quantifies that tradeoff; it does not tell us what a household can afford.

Pause and try 8

A loan starts at $2,000. Its payments total $2,180, including a smaller final payment. With no fees or new borrowing, how much interest was paid? Why is $2,180 not all interest?

6.9 Growing, unchanged, and decreasing are different outcomes

A payment does not automatically make a debt smaller. From $B_{\text{next}} - B = iB - M$, compare M with the interest iB. If the payment is larger, the balance falls. If it is equal, the balance stays the same. If it is smaller, the balance grows.



The first interest charge is $10 for all three plans. A $10 payment covers that interest exactly and leaves the original $1,000 unchanged; it does not pay the loan off.

With a fixed rate, fixed payment, and no new charges, a payment exceeding the initial monthly interest makes the balance decrease. Later interest is then no larger, so the same payment continues reducing the balance. In a fixed-rate model, this leads to eventual payoff; a calculator still needs enough periods to display the full path.

If a simulation stops at a chosen horizon, such as 1,200 months, an outstanding balance means "not paid within this horizon." It must not label the stopping month as payoff. At zero interest, a positive payment reduces the loan by a fixed amount each month; a zero payment leaves it unchanged.

Pause and try 9

An $8,000 balance has a 1.5% monthly rate. Classify payments of $100, $120, and $160 using the first interest charge.

6.10 A useful comparison includes the path, the totals, and the assumptions

A monthly payment, payoff time, and total interest answer different questions. For a borrower, the payment describes the current monthly commitment and the total describes its long-run modeled cost. A lender can use the same schedule to calculate payments received and interest collected. The underlying arithmetic is shared.

When comparing plans, name the starting principal, rate per period, timing of payments, rounding rule, and whether new purchases or fees are included. Then report what the model actually computes. Changing one assumption gives a new scenario, not a correction to a fact about every loan.

Example · A complete comparison statement

"For a $4,000 balance at 6% nominal annual interest with monthly updates, no additional charges, and cent-rounded monthly interest, $300 payments give payoff in month 14 and $150.01 total interest. With $150 payments, payoff is month 29 and interest is $303.84." This states both the result and the conditions supporting it.

1. Define the balance and the time unit.
2. Match the decimal rate to that unit.
3. Calculate one update by hand before using a formula or calculator.
4. Trace enough periods to answer the question.
5. Separate the ending balance, interest, payments, and total cost.
6. State a comparison and an assumption worth checking.

Prepare for Week 7

Week 7 compares costs, revenue, and profit for different possible quantities. The habit remains the same: define the input and units, name the outcome of interest, state the available choices, and interpret the result in context. Here the input is time; there it will often be quantity for one event or month.

Practice in the laboratory, compare the two repayment plans, and explore hypothetical compute growth. The discussion asks you to show and explain a payment schedule.

Two Further Applications

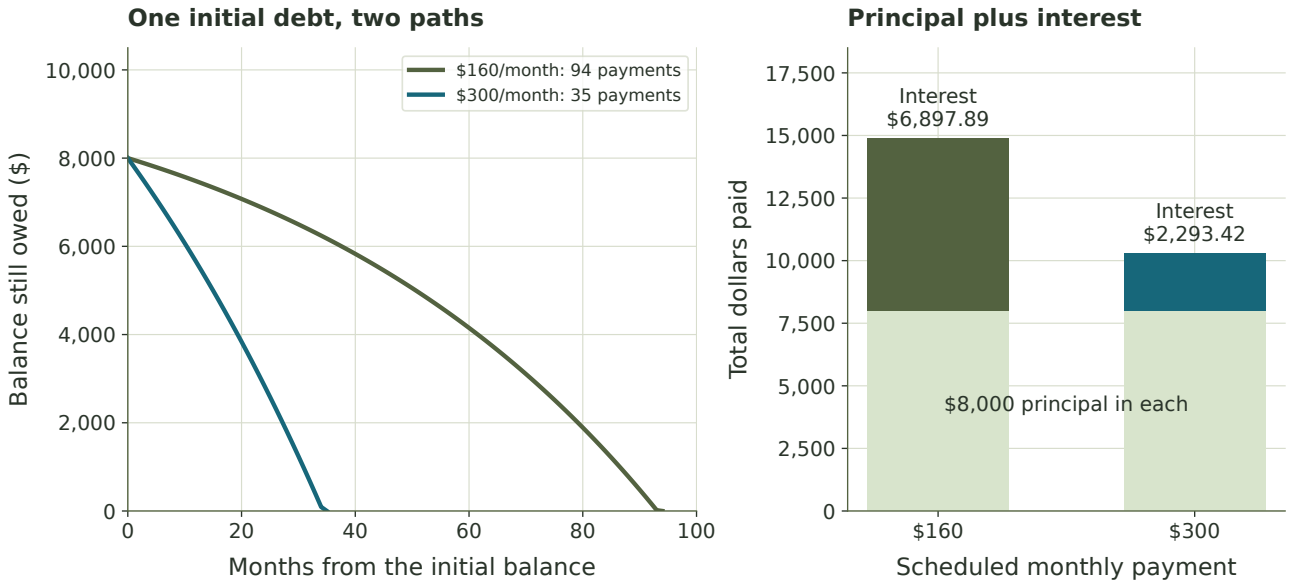
A fixed repayment comparison with full precision

The online application starts with an $8,000 balance, a nominal annual rate of 18%, and monthly interest $i = 0.015$. Interest is added first; fixed payments follow. There are no purchases, fees, or rate changes.

Both plans incur $120 first-month interest. A $160 payment reduces principal by $40 and ends at $7,960. A $300 payment reduces principal by $180 and ends at $7,820. In month 2, their interest charges are $119.40 and $117.30 because the starting balances now differ.

Regular payment	Payoff month	Total paid	Total interest
$160	94	$14,897.89	$6,897.89
$300	35	$10,293.42	$2,293.42

This table keeps full internal precision and rounds displayed money to cents. The actual final payments are approximately $17.89 and $93.42. The larger scheduled payment saves 59 months and about $4,604.46 of interest when the difference is calculated before display rounding. Subtracting the displayed rounded interest totals gives a one-cent difference.



The comparison describes the consequences of two stated schedules. Monthly affordability requires information beyond these model inputs. The CFPB's explanation of actual interest calculations also describes why a real card's daily-balance method can differ from our simplified monthly model.

A resource index can follow the same growth mathematics

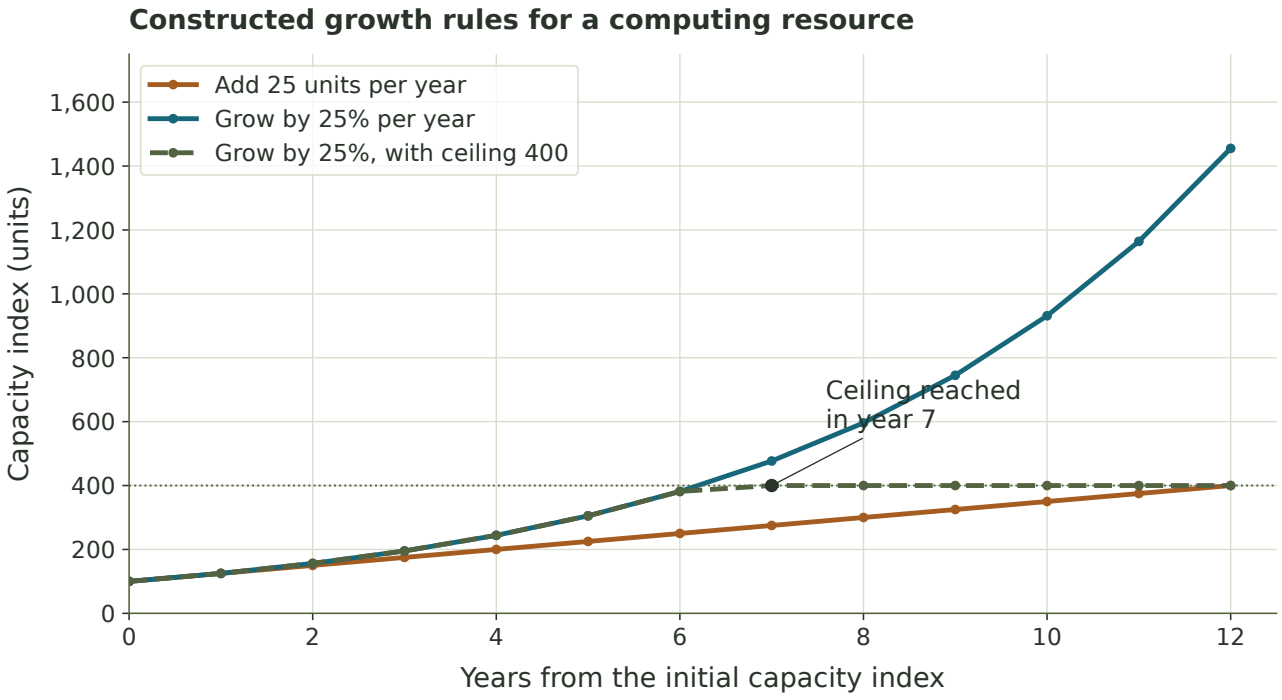
An invented computing-capacity index begins at 100 units. One plan adds 25 units each year: $A_k = 100 + 25k$. A second grows by 25%: $B_k = 100(1.25)^k$.

Both give 125 after one year. After two, the additive plan gives 150 but percentage growth gives 156.25. Its second increase is 31.25 because it acts on the updated base 125. These are resource-index values, not counts of whole machines or measurements of intelligence.

Suppose a third plan grows by 25% until a ceiling 400 is reached. Its rule is

$$L_0 = 100, \qquad L_{k+1} = \min(400, 1.25L_k).$$

Here min means choose the smaller value. At year 6, the limited and uncapped percentage plans both have about 381.47. At year 7, the uncapped value is about 476.84 while the limited rule selects 400.



The ceiling changes the rule and thus the trajectory. It is an explicit hypothetical policy, not a forecast about any real AI operation. Learning, reliability, energy, and useful service each require suitable measures and observations.

Chapter 6 Review

Show units and at least one substituted expression or update. For savings formulas, keep precision until the final displayed answer. For loan tables, use monthly cent rounding unless the question explicitly says full precision. Assume no fees or new borrowing unless stated.

1. A \$5,000 account earns 4% simple interest annually. Write its balance formula, give balances for years 0 through 5, and find the time to \$6,000. What is the fixed yearly interest? Also recover the simple annual rate when \$500 becomes \$560 in four years.

2. The same \$5,000 instead earns 4% compounded annually. Give balances for years 0 through 5, compare the interest in year 1 with year 5, and find how much more it holds after five years than the simple model. Explain the source of the extra gain.

3. For \$1,000 at a 6% nominal annual rate over ten years, compare annual, quarterly, monthly, and daily compounding. For daily, use 365 equal periods per year. Report balances, interest earned, and the monthly decimal rate and update count.

4. A 6% nominal annual rate is compounded quarterly for two years. Find the rate per quarter, the number of factors, and the final balance on \$1,000. Separately, find the effective annual yield for a 3% nominal annual rate compounded monthly, as both a decimal and a percentage.

5. Use the Rule of 72 to estimate (a) doubling at 12% annually, (b) doubling at 7%, (c) quadrupling at 7%, and (d) the annual rate for doubling in six years. Explain why these are estimates under a continuing compound-growth rule.

6. A hypothetical unpaid \$30,000 balance grows by 6% compounded annually, with no other charges. Estimate its balance after 12 years using the Rule of 72, then compute the compound value. Check the factors at years 11 and 12 to locate the first completed annual update reaching twice the original amount. Do not assume a real loan necessarily follows this no-payment rule.

7. A \$2,500 loan has monthly rate 0.01 and payment \$120. Build the first four monthly rows, with interest, actual payment, principal reduction, and ending balance. Compare a rough estimate based on a constant \$95 monthly principal reduction with the actual schedule's payoff in month 24. Why does the rough estimate differ?

8. For the reading's \$4,000 loan at 6% nominal annual interest and \$200 monthly payment, verify the first two rows and the last-payment calculation. Use monthly cent rounding. Find total paid and total interest. Why would multiplying 22 by \$200 overstate the amount paid?

9. Two $5,000 loans use a 6% nominal annual rate with monthly payments $150 and $250. Compare first-month interest, first ending balances, and second-month interest, using cent rounding. Predict which fixed schedule has lower total interest and explain the balance mechanism.

10. A $1,000 debt has monthly rate 1%. Classify fixed payments $5, $10, and $50. Separately, how long does $1,000 at zero interest take to repay with $100 monthly payments? What if its payment is zero? What if the starting debt is zero?

11. Account A starts at $1,000 with 5% simple annual interest. Account B starts at $1,000 with 4% annual compounding. Compare balances at years 5, 10, 20, and 30. Check years 11 and 12 to find when B first overtakes A at a completed annual update. Why can B eventually lead despite its smaller percentage rate?

12. A hypothetical unpaid $3,000 balance has a 24% nominal annual rate divided across monthly updates. Find the monthly percentage. Apply the Rule of 72 in *monthly periods* and compute balances after 12 and 36 months. Should the 12-month result already be double? Explain.

13. In the $8,000 application, use full internal precision to compare $160 and $300 payments at nominal annual rate 18%. Give payoff months, final actual payments, total paid, and total interest. Explain the first-month agreement and later divergence. Why can subtracting rounded displayed totals differ by one cent from rounding their unrounded difference?

14. Optional horizon investigation: a $200,000 loan uses nominal annual rate 4.5%, monthly interest, and a fixed payment rounded to $1,013.37. Compute the budget estimate of 360 such payments, its amount above principal, and that excess as a percentage of principal. Then use cent rounding and a 360-month horizon in the explorer. Does the balance actually reach zero? Explain the residual and how a simulator must report it.

15. Compare the reading's $4,000, 6%, $200-payment schedule using cent-rounded monthly interest versus full internal precision. Give payoff month, final payment, and total interest under each convention. What explains a small difference? What information should accompany a reported total?

16. A hypothetical capacity index begins at 100. Compare adding 25 per year, growing 25% per year, and growing 25% with a ceiling 400. Find all three values at year 2 and year 12, and locate the limited plan's first departure from the uncapped plan. Do equal endpoints establish identical paths? Name an important outcome this resource index does not measure.

Solutions and Reasoning

Short check 1. The decimal rate is 0.04, interest is 0.04(500)=$20, the factor is 1.04, and the new balance is $520.

Short check 2. The increase is 900(0.04)=$36 per year. After three years, the total is 900+3(36)=$1,008. The $108 gain is interest, not the total balance.

Short check 3. The balances are 300(1.10)=$330 and 330(1.10)=$363. Simple interest gives 300+2(30)=$360. The extra $3 comes from interest on the first year's $30 interest.

Short check 4. The monthly rate is 0.12/12=0.01, or 1%. There are 12(2)=24 updates, so the growth factor over the full time is (1.01)^24.

Short check 5. At 8%, one doubling takes about 72/8=9 years. At 6%, one doubling takes about 12 years, so quadrupling takes about 24 years if the rate and compounding assumptions continue.

Short check 6. Interest is 0.02(500)=$10. The payment reduces principal by 60−10=$50. Ending balance is 500+10−60=$450.

Short check 7. Principal reduction is 200−16.37=$183.63. Under a fixed positive rate, reducing the balance makes the next interest charge smaller; the rate itself does not need to change.

Short check 8. Total interest is 2180−2000=$180. The other $2,000 repays the principal originally borrowed.

Short check 9. The first interest is 8000(0.015)=$120. A $100 payment leaves the balance growing, $120 leaves it unchanged, and $160 reduces it by $40 in month 1, under the fixed-rate monthly model.

Chapter review: solutions

1. $A(t) = 5000(1 + 0.04t) = 5000 + 200t$. Balances are 5000, 5200, 5400, 5600, 5800, 6000 dollars, so \$6,000 is reached at year 5. Each year adds \$200 based on the original principal. For \$500 growing to \$560 in four simple-interest years, $r = (560/500 - 1)/4 = 0.03$, or 3% annually.

2. $B_t = 5000(1.04)^t$ gives 5000.00, 5200.00, 5408.00, 5624.32, 5849.29, 6083.26 dollars. Year 1 earns \$200; year 5 earns approximately \$233.97. The extra balance over the simple model is approximately \$83.26. Earlier interest is included in later percentage calculations; the rate itself stays at 4%.

3. Final balances are approximately 1790.85 annually, 1814.02 quarterly, 1819.40 monthly, and 1822.03 daily. Subtract 1000 to get interest 790.85, 814.02, 819.40, and 822.03 dollars. The monthly decimal rate is 0.06/12=0.005 and the count is 120. The comparison fixes the nominal annual rate, starting amount, and duration.

4. Quarterly rate is 0.06/4=0.015, or 1.5%, with 8 factors. $1000(1.015)^8 \approx 1126.49$. The effective annual yield at 3% nominal with monthly compounding is $(1 + 0.03/12)^{12} - 1 \approx 0.030416$, or about 3.0416%. Multiply the decimal yield by 100 to express it as a percentage.

5. (a) $72/12 = 6$ years. (b) $72/7 \approx 10.29$ years. (c) About $2(72/7) \approx 20.57$ years. (d) About 12% annually. The Rule of 72 is an approximation for continuing percentage growth, not an equality identifying an exact balance on a particular crediting date.

6. One estimated doubling gives approximately \$60,000. Direct calculation gives $30000(1.06)^{12} \approx 60365.89$. Factors at years 11 and 12 are about 1.898 and 2.012. Thus the first completed annual update reaching twice the start is year 12. The small excess over two is consistent with the rule being an estimate.

7. Interest/end-balance pairs for months 1–4 are (25.00,2405.00), (24.05,2309.05), (23.09,2212.14), and (22.12,2114.26), all dollars. Subtract each interest from 120 to get principal reductions 95.00, 95.95, 96.91, and 97.88. A constant \$95 reduction gives $2500/95 \approx 26.32$ months, or 27 whole payments to clear the amount under that simplified rule. The actual schedule finishes in 24 months because the declining balance reduces interest and increases the principal part of later payments.

8. Month 1 is interest 20.00, payment 200, end 3820.00. Month 2 is interest 19.10, payment 200, end 3639.10. After 21 payments, 24.87 remains. Month 22 adds 0.12 interest, so actual payment is 24.99 and end is zero. Total paid is $21(200) + 24.99 = 4224.99$; total interest is 224.99. Twenty-two full payments would count \$175.01 that is not owed under this schedule.

9. Both first interest charges are \$25. With a \$150 payment the first end is \$4,875; with \$250 it is \$4,775. Second interest is \$24.38 versus \$23.88 after cent rounding. At the same fixed rate and without new charges, the larger payment leaves a smaller base for later interest, shortening the schedule and reducing total interest. The extra monthly commitment is a separate consideration.

10. The first interest charge is \$10. Payment 5 leaves growing debt; 10 keeps it constant; 50 reduces it and eventually repays it under the fixed rule. At zero interest, ten \$100 payments repay \$1,000. With zero payment it stays owed unchanged. A zero starting debt needs no payments and incurs no interest in this model.

11. A's balances are 1250, 1500, 2000, 2500 at years 5, 10, 20, 30. B's are approximately 1216.65, 1480.24, 2191.12, 3243.40. At year 11, A is 1550 while B is about 1539.45; at year 12, A is 1600 while B is about 1601.03. B first overtakes at year 12 among completed annual updates. Its dollar increases grow with its base, while A continues adding the fixed \$50.

12. Monthly rate is 0.24/12=0.02, or 2%. The Rule of 72 estimates $72/2 = 36$ *months* per doubling. After 12 months, $3000(1.02)^{12} \approx 3804.73$; after 36, $3000(1.02)^{36} \approx 6119.66$. The first value is not yet near a full doubling because only one of the estimated three years has passed. The approximation is consistent with the roughly doubled 36-month value.

13. The \$160 plan finishes in month 94 with a final payment about \$17.89, total paid \$14,897.89, and total interest \$6,897.89. The \$300 plan finishes in month 35 with final payment about \$93.42, total paid \$10,293.42, and interest \$2,293.42. Both first interest charges are \$120, calculated before payment. Their differing balances then produce different charges. Subtracting full-precision interest totals gives savings about \$4,604.46; subtracting the separately rounded displays gives \$4,604.47 because rounding is applied at a different stage.

14. The budget estimate is $360(1013.37) = 364813.20$. Its excess above principal is 164813.20, or 82.4066% of 200000. This total is a calculation about 360 scheduled amounts, not proof of exact payoff. With monthly cent rounding, \$0.63 still remains after payment 360. A fixed-payment simulation reports an outstanding balance at that horizon. Under the same capped-payment rule, a 361st actual payment of \$0.63 clears it; total paid is then \$364,813.83. An adjusted final closing payment would be another explicit schedule choice.

15. Both finish in month 22. Monthly cent rounding gives final payment \$24.99 and total interest \$224.99. Keeping full internal precision gives a final payment about \$25.00 and total interest about \$225.00. The one-cent displayed difference arises from the stages at which rounding occurs. Report the rate, timing, and rounding convention with the totals.

16. At year 2, additive gives 150; uncapped and limited percentage growth give 156.25. At year 12, additive gives 400, uncapped growth about 1455.19, and limited growth 400. The limited plan first differs in year 7: its cap gives 400 instead of the uncapped 476.84. Matching year-12 endpoints for the additive and limited plans do not imply matching earlier values. Learning, reliability, useful service, and energy impact require separate definitions and observations.

Further reading

Investor.gov: What is compound interest? introduces compounding and the Rule of 72. The Investor.gov compound-interest calculator can also include later contributions; this chapter's basic growth comparison uses none. The CFPB source linked in the loan application explains daily-balance calculations that can differ from our classroom monthly model.